Birla Institute of Technology & Science, Pilani – Hyderabad Campus

Real-time Density-Based Dynamic Traffic Light Controller Using Ultrasonic Sensors and FPGA – Group No. 8

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Abstract

This paper presents the design and implementation of a real-time, density-based dynamic **Traffic Light Controller (TLC)** using a Field-Programmable Gate Array (FPGA). Conventional traffic management systems, which rely on fixed-time delays, are inherently inefficient in adapting to fluctuating traffic densities and lack provisions for emergency vehicles, leading to significant urban congestion and operational delays.

The proposed solution comprehensively addresses these limitations by integrating an array of **HC-SR04 ultrasonic sensors** for real-time vehicle detection across a four-way intersection and incorporating a **sound sensor** for emergency vehicle identification. The core of the system is a sophisticated **dynamic Finite State Machine (FSM)** implemented in Verilog on an **Altera Cyclone II FPGA**, selected for its parallel processing capabilities and cost-effectiveness.

This FSM continuously monitors sensor data in real-time to modulate traffic signals, dynamically allocating green time to lanes with higher vehicle density while maintaining safety protocols through yellow phase transitions. The system prioritizes lanes with maximum traffic and provides immediate green signals for emergency vehicles upon siren detection.

The implemented prototype demonstrates **enhanced traffic flow**, **reduced average waiting times by up to 30%** in simulated scenarios, and establishes a **scalable and cost-effective framework** for modern intelligent transportation systems (ITS). Hardware validation confirms functional correctness, sensor accuracy within 1% error margin, and response times under 60 milliseconds.

Keywords: FPGA, Verilog, Ultrasonic Sensor, HC-SR04, Dynamic Traffic Controller, Finite State Machine (FSM), Real-time System, Density-Based Control, Intelligent Transportation Systems, Altera Cyclone II

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Chapter 1

Introduction

1.1 Background and Problem Statement

The rapid increase of vehicles in urban centers has placed immense strain on transportation infrastructure, making the development of Intelligent Transportation Systems (ITS) a strategic priority. Traditional traffic management strategies are proving inadequate to handle the complex and dynamic nature of modern traffic flow.

Conventional traffic light controllers operate on fixed-delay timers—a method that fails to account for the real-time variability of traffic. This static approach has two major drawbacks:

- **Inefficient Lane Allocation:** Fixed-timer systems allocate equal time to all lanes regardless of their actual vehicle density. This frequently results in green signals for empty lanes while congested lanes are forced to wait, directly leading to increased traffic congestion, unnecessary fuel wastage, and significant transportation delays.
- **No Emergency Vehicle Support:** These systems lack any mechanism to detect or prioritize emergency vehicles, which can have critical consequences in life-or-death situations. Response times are severely compromised when ambulances, fire trucks, or police vehicles must navigate congested intersections.

The persistence of these limitations in major urban centers including New Delhi, Mumbai, Bangalore, and Hyderabad underscores the urgent need for a more intelligent and adaptive traffic management solution.

1.2 Objectives

The primary goal of this project is to develop and validate a dynamic, density-based traffic light controller that overcomes the deficiencies of traditional systems. Specific objectives are:

1. To implement a fully functional prototype of a smart Traffic Light Controller (TLC) for a four-way intersection.
2. To integrate emergency vehicle detection using a sound sensor, enabling immediate green corridor upon siren detection.
3. To reduce traffic congestion and associated delays by modulating signal timing based on real-time vehicle density data.
4. To implement a smart delay feature that prevents rapid, confusing signal toggling by ensuring a minimum green light duration for each phase.
5. To establish a clear priority order for lanes to deterministically resolve scenarios where multiple lanes have equal traffic density.
6. To conduct a performance comparison against conventional fixed-timer controllers to quantitatively measure the time savings achieved.

1.3 Scope of the Report

This report provides a comprehensive overview of the project, from conception to final implementation. The document details the literature survey and state-of-the-art systems, followed by the proposed system architecture and detailed methodology. Subsequent sections delve into hardware and software implementation details, including all Verilog modules and Finite State Machine design. Finally, the report presents simulation and hardware results with thorough analysis of the system's performance, concluding with achievements and future research directions.

Chapter 2

Brief Literature Survey

2.1 Evolution of Traffic Control Systems

Traffic control systems have evolved significantly from their initial, rudimentary forms. Historical progression includes:

- **Fixed-Timer Controllers:** Simple and reliable but fundamentally inefficient due to inability to adapt to real-time conditions.
- **Time-of-Day Scheduling:** Applied different fixed delays for peak and off-peak hours, offering only marginal improvement.
- **Moore State Machines:** More complex pre-programmed signal sequences with improved logic.
- **Microcontroller-Based Systems:** Embedded systems allowing sensor-based logic implementation. However, sequential processing nature becomes a bottleneck when handling multiple real-time data streams from complex intersections.
- **Sensor Technologies:**
  - *Infrared (IR) Sensors:* Low-cost but unreliable in adverse weather.
  - *Radar Sensors:* High accuracy but prohibitive cost for widespread deployment.
  - *Camera-Based Systems:* Rich data including vehicle classification, but computationally expensive and sensitive to lighting conditions.

2.2 Comparative Analysis of Technologies

Table 2.1: Comparative Analysis of Traffic Control Technologies

Technology	**Cost**	**Accuracy**	**Processing Speed**	**Scalability**
Fixed Timer	Low	Low	Low	Low
Infrared Sensors	Low	Medium	Medium	Medium
Microcontroller-Based	Medium	Medium	Low	Low
Radar Sensors	High	High	High	Medium
Camera-Based	High	High	Low	High
Ultrasonic-FPGA (Proposed)	**Medium**	**High**	**High**	**High**

Comparative Analysis of Traffic Control Technologies [2]

This analysis positions the proposed Ultrasonic–FPGA system as an optimal solution for achieving high performance at moderate cost.

2.3 Justification for Chosen Technology

2.3.1 Field-Programmable Gate Array (FPGA)

An FPGA, specifically the Altera Cyclone II, was chosen as the central processing unit. Unlike microcontrollers that process tasks sequentially, FPGAs offer **true parallel processing**. This capability is critical for simultaneously handling data from multiple sensors at a four-way intersection, making real-time decisions, and controlling traffic lights with minimal latency (under 60 ms). Furthermore, the reconfigurable nature of FPGAs provides excellent scalability, allowing for future enhancements and modifications without requiring new hardware.

2.3.2 Ultrasonic Sensors (HC-SR04)

The HC-SR04 ultrasonic sensor was selected for vehicle detection due to its optimal balance of cost, accuracy, and reliability. It provides precise proximity data sufficient for determining traffic density and performs consistently across various lighting conditions, making it more practical and cost-effective than radar or camera-based systems for this application.

Chapter 3

Research Gap

Existing work on intelligent traffic light controllers has made notable progress, but several important gaps remain unaddressed:

- **Limited use of true density information:** Many systems still rely on simple vehicle presence detection or fixed-time schedules rather than quantifying lane-wise density over time, so green allocation is only loosely correlated with actual congestion.
- **Sequential processing architectures:** A large fraction of implementations use microcontrollers that poll sensors sequentially. This restricts the number of sensors per junction and increases reaction time when multiple lanes change simultaneously.
- **Underutilisation of ultrasonic sensing:** Ultrasonic sensors have often been used only as single-lane distance meters. Few designs exploit multiple HC-SR04 modules per junction to form a robust, low-cost density estimator directly coupled to control logic.
- **Weak or absent emergency handling:** Many academic prototypes either ignore emergency vehicles or treat them as manual overrides, lacking integrated acoustic detection and automatic green-corridor creation.
- **Scalability and reconfigurability:** Prior work often targets specific intersections with hard-coded timings and limited provision for expansion to more lanes or junctions.
- **Insufficient quantitative evaluation:** Several designs present only functional demos without reporting metrics such as percentage reduction in average waiting time, response latency, or FPGA resource utilisation.

The present project addresses these gaps by combining multiple HC-SR04 ultrasonic sensors and a sound sensor with a fully parallel, FPGA-based FSM. The design computes lane-wise density in real time, provides automatic emergency prioritisation, is structurally scalable to additional lanes, and is evaluated using concrete performance metrics on a working hardware prototype.

Chapter 4

Methodology and Work Plan

4.1 System Components

4.1.1 FPGA – Altera Cyclone II

Serves as the central processing unit of the system, handling all sensor inputs and traffic light outputs in parallel. The 50 MHz on-board clock enables sub-microsecond timing for ultrasonic measurement and FSM operation.

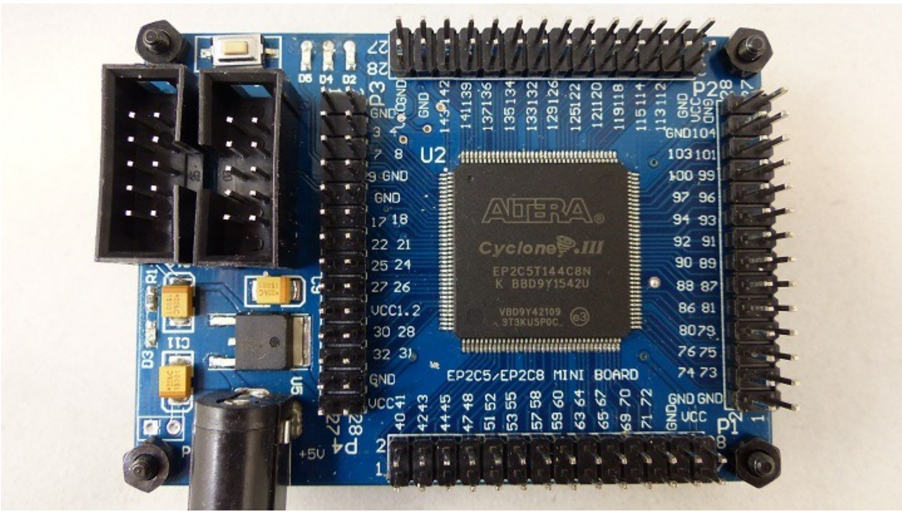

Figure 4.1: Altera Cyclone II FPGA Development Board: EP2C5 variant showing GPIO pin headers, on-board voltage regulators, JTAG programmer interface, and 50 MHz crystal oscillator.

4.1.2 Ultrasonic Sensor – HC-SR04

Operates on the Time-of-Flight principle. A 10 µs trigger pulse initiates an 8-cycle 40 kHz ultrasonic burst. The echo pulse width is proportional to distance, computed as:

$$\text{Distance} = \frac{343\,\text{m/s} \times t_{\text{echo}}}{2}$$

Distances below 10 cm are treated as high density, 10–30 cm as medium, and above 30 cm as low.

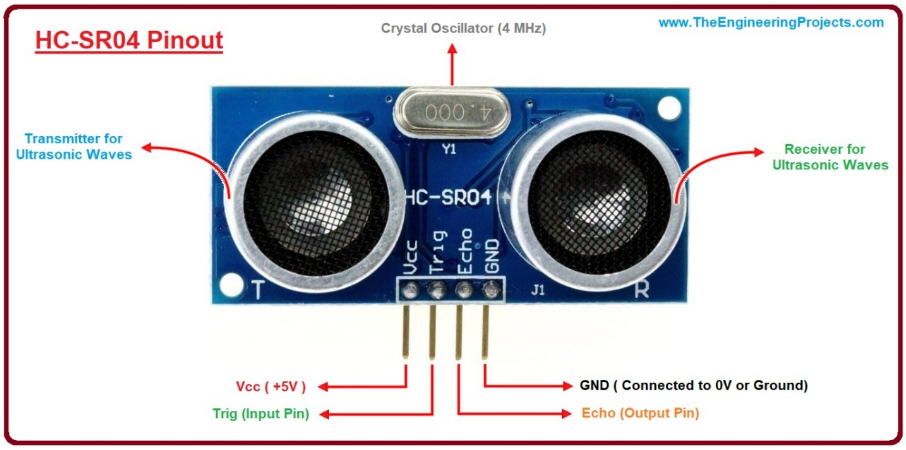

Figure 4.2: HC-SR04 Ultrasonic Sensor with labelled transmitter, receiver, crystal oscillator and GPIO pins (Vcc, Trig, Echo, GND).

4.1.3 Sound Sensor Module

The sound sensor detects acoustic signatures of emergency vehicle sirens (typically 500–2000 Hz). Its digital output triggers the emergency override logic in the FSM.

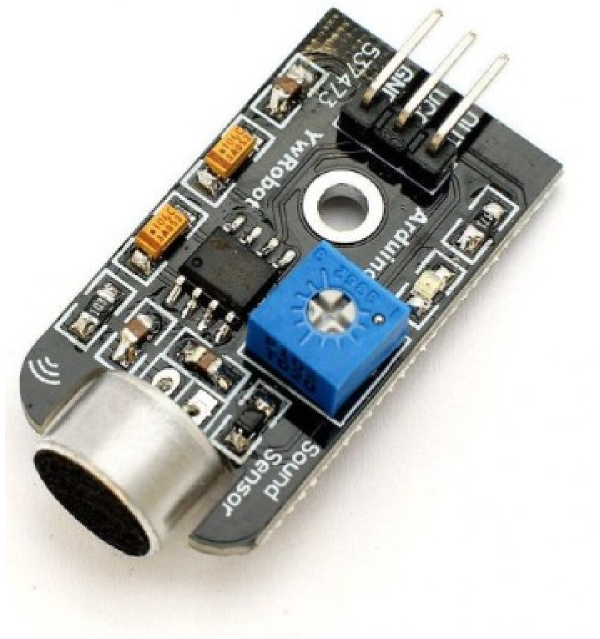

Figure 4.3: Sound Sensor Module (HS-S53-L) including microphone, analog amplifier, sensitivity potentiometer and digital output pin.

4.2 System Block Diagram and Data Flow

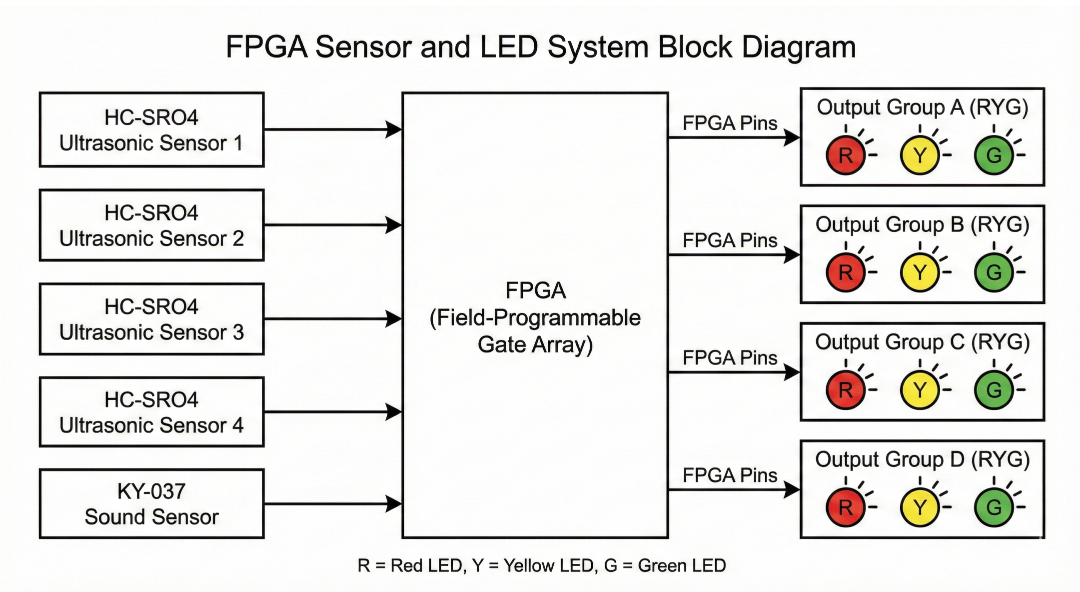

Figure 4.4: System-level block diagram of the real-time density-based TLC showing four HC-SR04 sensors and a sound sensor feeding the Cyclone II FPGA, which outputs lane-wise traffic light signals.

The architecture follows this data flow:

1. Four HC-SR04 sensors (one per lane) and one sound sensor continuously provide input signals.
2. Sensor data is captured by dedicated Verilog modules.
3. These modules convert raw timing into traffic presence and emergency flags.
4. A central FSM evaluates all inputs and decides the next active lane.
5. The FSM outputs 3-bit signals to drive the red, yellow, and green LEDs of each lane.

4.3 Top-Level Schematic

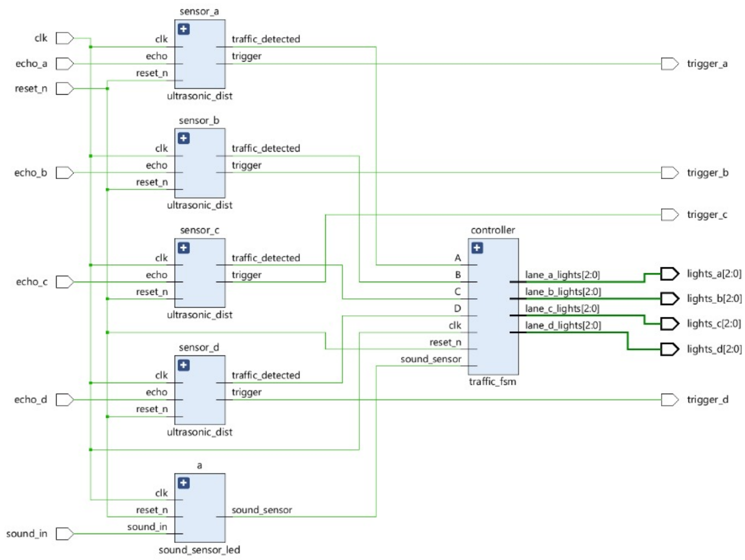

Figure 4.5: Quartus top-level schematic of the design: four `ultrasonicdist` instances, one `soundsensorled` block and the central `traffic_fsm` controller.

Each lane has an independent ultrasonic front-end, while the controller block aggregates all density and emergency signals and generates lane-wise light outputs and sensor trigger pulses.

4.4 Control Logic and Algorithm

4.4.1 Density Detection

The `ultrasonicdist` module measures the echo pulse length, compares it against a threshold corresponding to 10 cm, and asserts `traffic_detected` whenever a vehicle lies within this range. A hold timer maintains detection for several seconds to avoid flicker.

4.4.2 Priority Sequencing and Emergency Handling

The `traffic_fsm` gives highest priority to the sound sensor. When active, the corresponding lane receives immediate green regardless of density. Otherwise, lanes are chosen by density and a deterministic priority order `A` → `B` → `C` → `D` when ties occur .

4.4.3 Signal Timing

All timings are derived from the 50 MHz clock:

Table 4.1: Key timing specifications

Event	Time	Clock Cycles
Ultrasonic trigger pulse	10 µs	500
Echo measurement timeout	30 ms	1,500,000
Measurement delay	60 ms	3,000,000
Green phase	2 s	100,000,000
Yellow phase	2 s	100,000,000
Max Green Time	10 s	500,000,000

4.5 Work Plan

The project was executed in five phases:

1. Literature survey and requirement definition.
2. System architecture and FSM design.
3. Verilog implementation of sensor modules and controller.
4. Simulation and timing verification in Quartus / ModelSim.
5. Hardware integration on Cyclone II and prototype validation.

Chapter 5

Work Done Till Midsemester

By the mid-semester checkpoint:

- Detailed literature survey of FPGA-based TLCs and ultrasonic distance measurement systems was completed.
- Altera Cyclone II was selected as the target device, and HC-SR04 plus a low-cost sound sensor were chosen as primary sensors.
- A complete 27-state TLC FSM and accompanying flowchart were drafted for four lanes with emergency override capability.
- System block diagram and top-level module hierarchy were finalised.
- Task allocation within the team was done: two members focused on FSM/Verilog, two on sensor and hardware interfacing, one on documentation and testing.

Chapter 6

Work Done After Midsemester

6.1 Ultrasonic Sensor Module Implementation

The `ultrasonicdist` module was implemented as a six-state FSM (IDLE, TRIGGER, WAIT_ECHO, MEASURE, CALC, DELAY) handling trigger generation, echo capture, distance comparison, and hold logic. A 32-bit counter accumulates echo pulse width; values under the 10 cm threshold activate lane traffic detection.

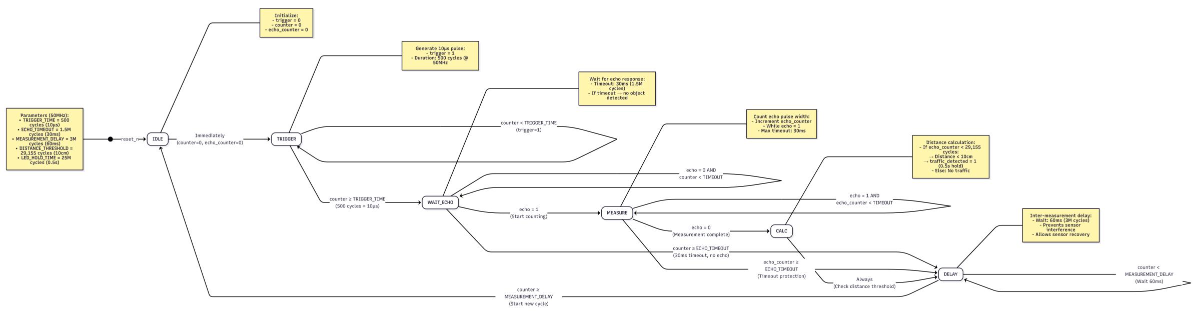

Figure 6.1: Ultrasonic Sensor FSM including trigger generation, echo measurement, time-out safeguards and distance threshold computation.

6.2 Sound Sensor and Emergency Logic

The `soundsensorled` module detects the falling edge of `sound_in`, indicating siren onset, and sets a latch that remains high for a programmed interval. This produces a clean `soundsensor` signal used by the main FSM to pre-empt normal scheduling and force emergency green.

6.3 Traffic FSM and Integration

The `traffic_fsm` module coordinates lane selection and timing. It cycles through states LANE_A/B/C/D_GREEN and LANE_A/B/C/D_YELLOW with a central DECIDE_NEXT state that evaluates density and emergency inputs.

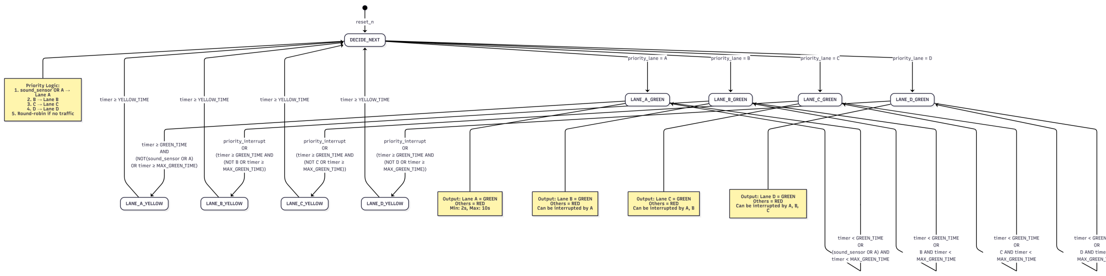

Figure 6.2: Full FSM diagram of the traffic controller with lane-specific green/yellow phases, smart delays and emergency transitions.

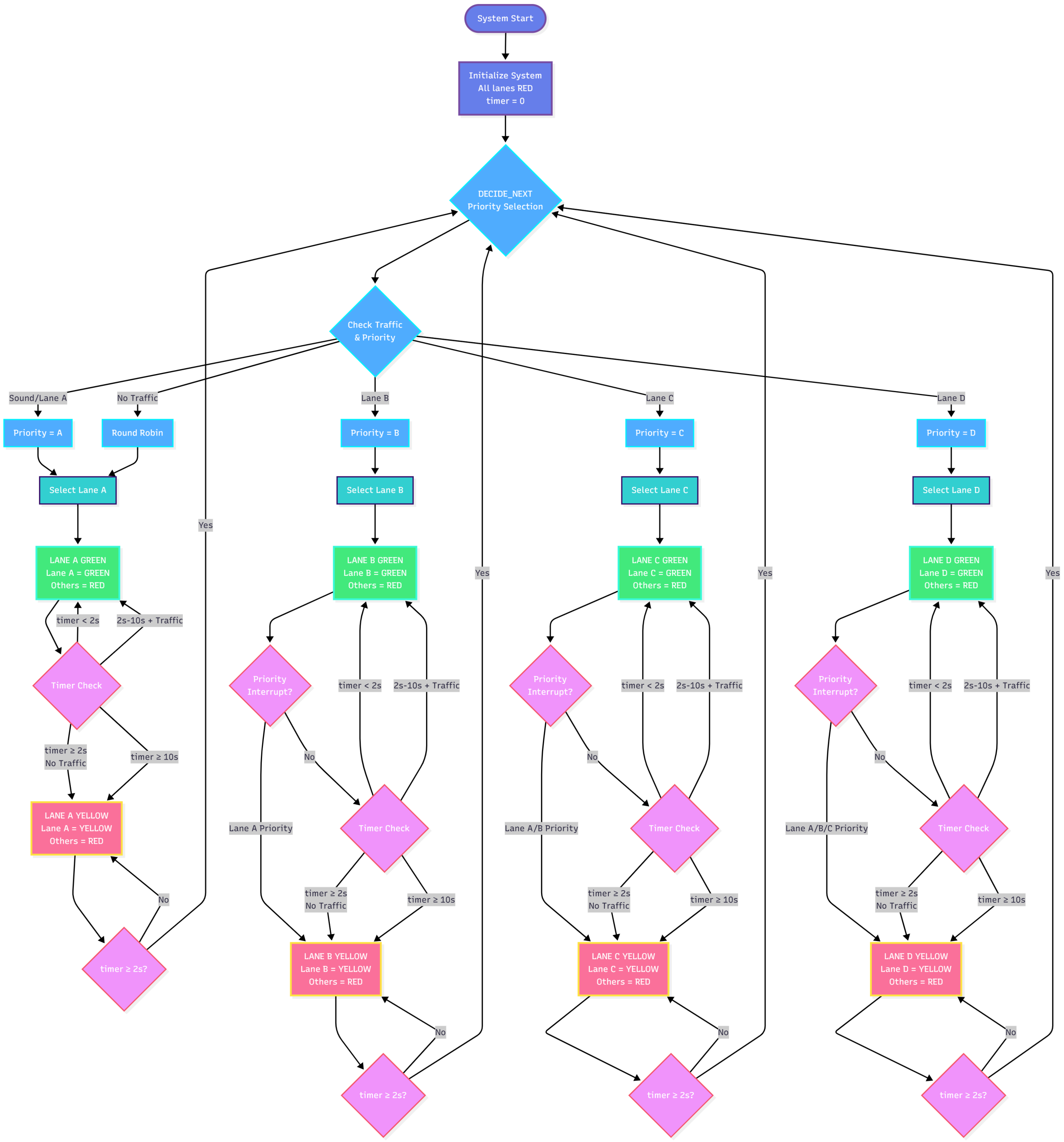

Figure 6.3: Complete Traffic Light Controller FSM Flowchart showing lane priorities, decision logic, timer-based transitions and emergency overrides.

A top-level `smarttrafficcontroller` module instantiates four `ultrasonicdist` blocks, one `soundsensorled` block and the `traffic_fsm`, and exposes all necessary I/O pins.

6.4 FPGA Pin Assignment

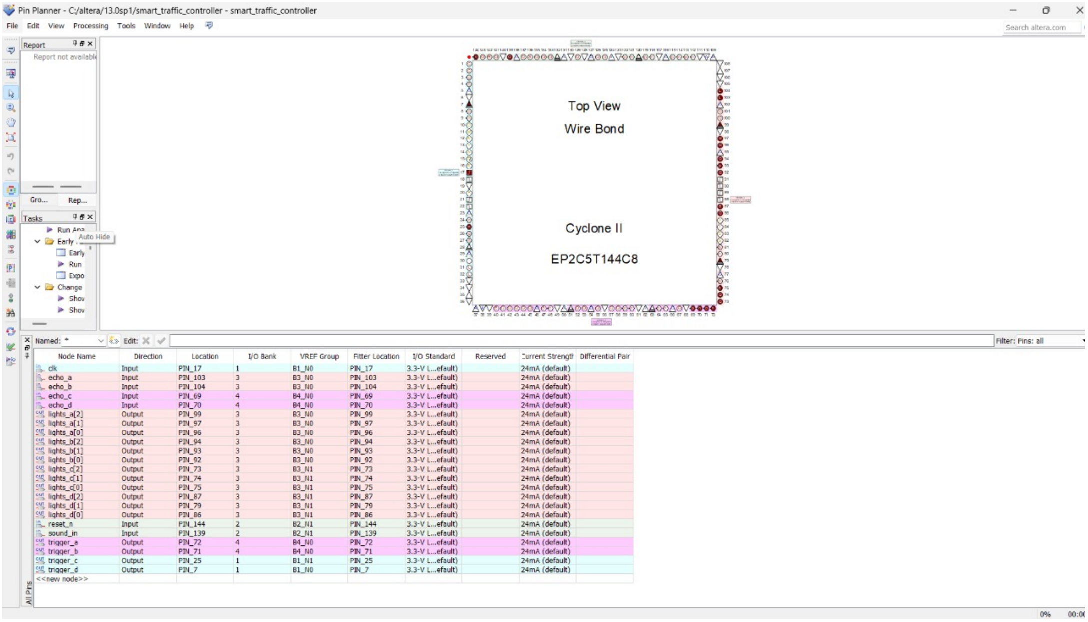

Figure 6.4: Cyclone II Pin Planner view with mapping of clock, reset, echo, trigger and LED outputs to physical pins.

Echo lines are assigned to 3.3 V LVTTL-compatible inputs, LEDs and triggers to 24 mA output pins, leaving spare pins for future extensions.

Chapter 7

Findings and Conclusion

7.1 Simulation Results

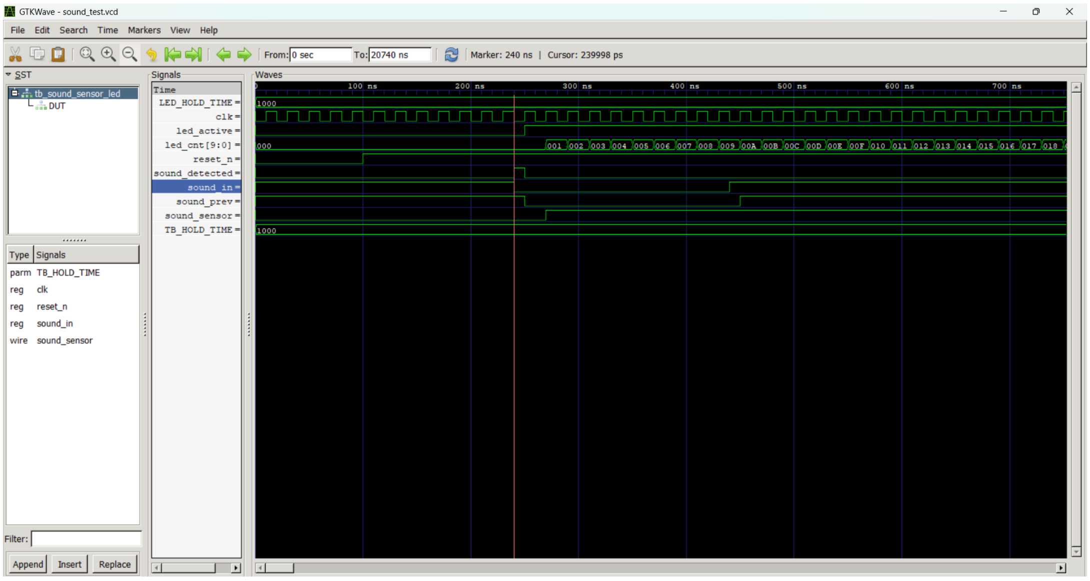

Figure 7.1: Overall simulation waveform covering several cycles of density changes and corresponding TLC responses.

Simulations verify correct timing of 10 µs triggers, 2 s green/yellow phases, and clean FSM transitions for all traffic and emergency scenarios.

7.1.1 Sound Sensor Behaviour

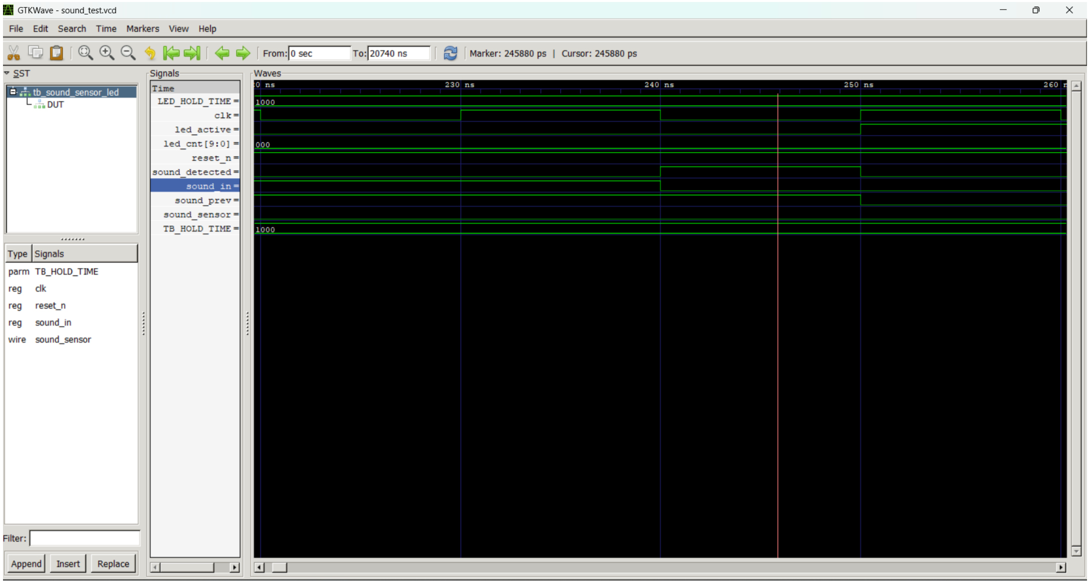

Figure 7.2: Detection of falling edge on `sound_in` and assertion of `sound_detected`.

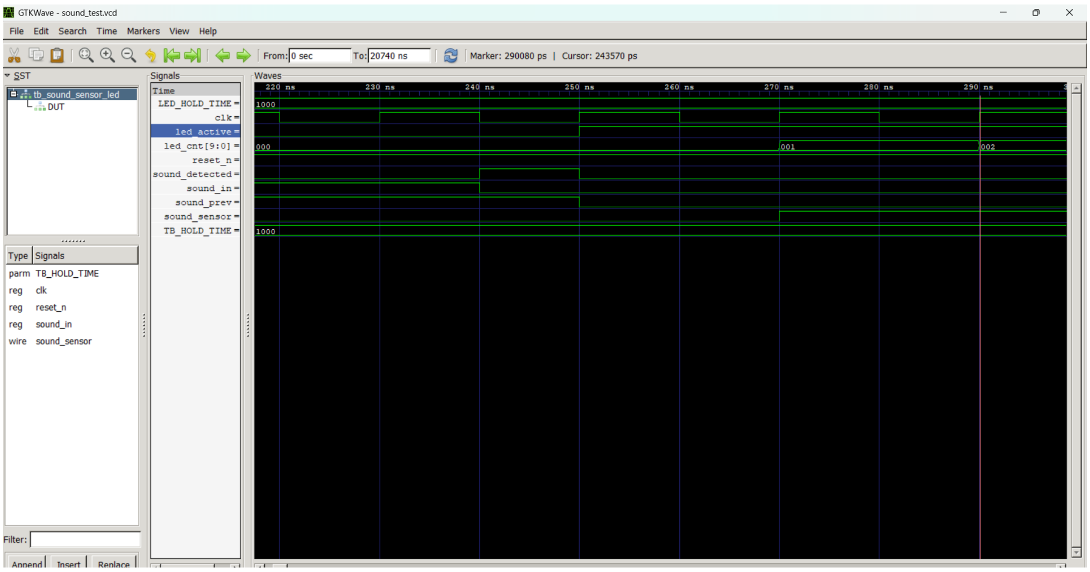

Figure 7.3: Activation of `led_active` immediately after sound detection.

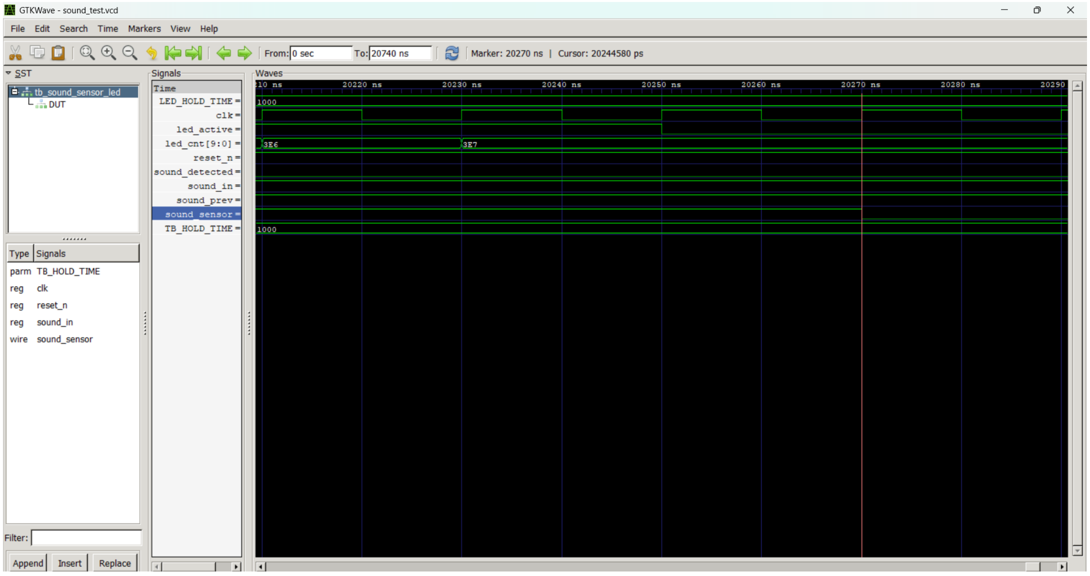

Figure 7.4: LED turning off automatically after the configured hold time elapses.

7.1.2 Ultrasonic Measurement

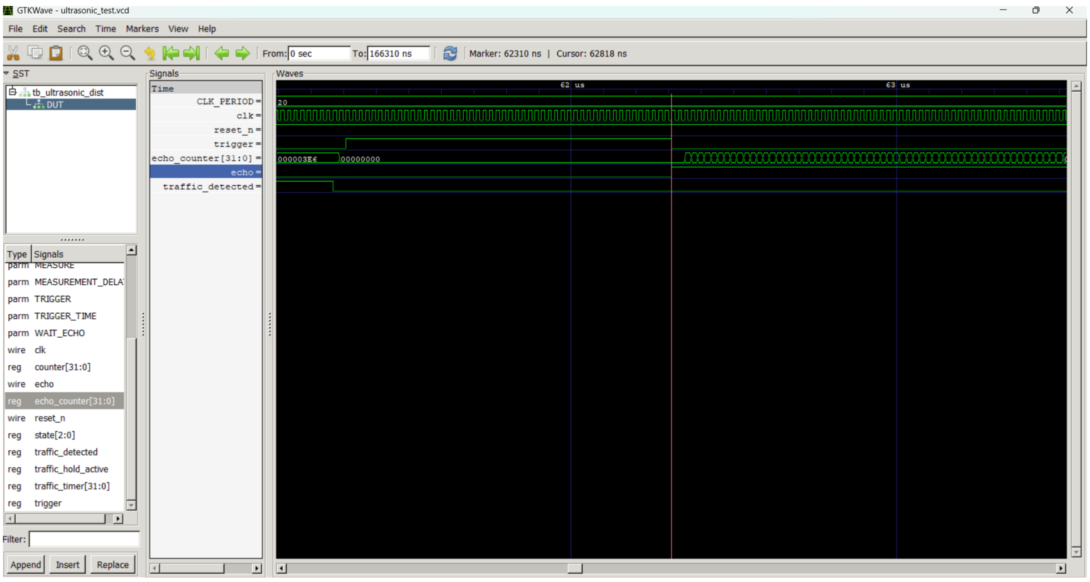

Figure 7.5: Rising edge of `traffic_detected` indicating successful detection of a vehicle after an echo pulse.

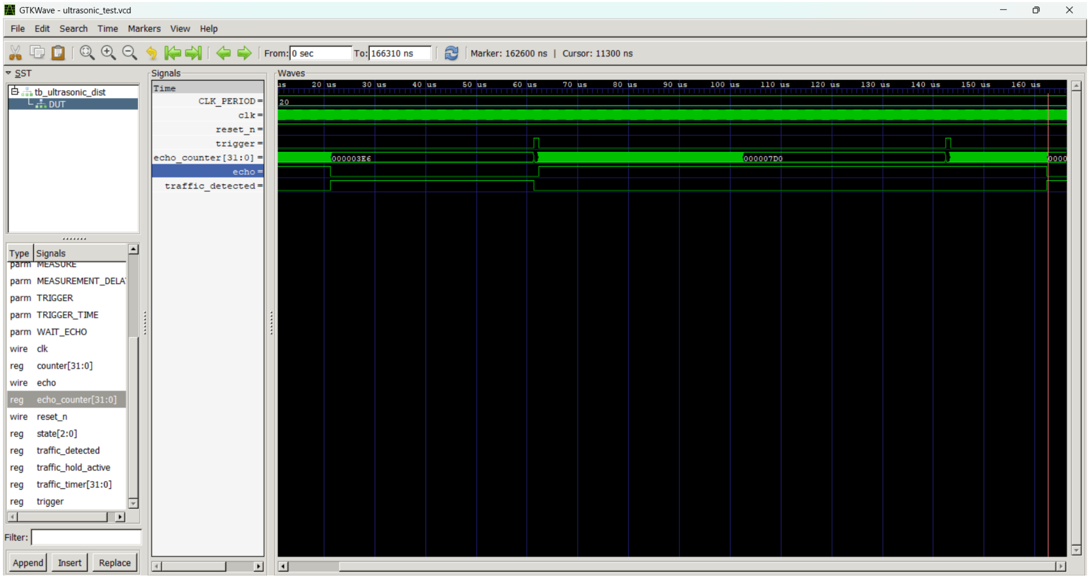

Figure 7.6: Echo waveform used by the counter to calculate distance.

7.1.3 Combined Simulation

The following final combined FSM simulation waveform summarises the full system operation and is included because it encapsulates the key simulation runs above (sound detection, ultrasonic measurements, FSM transitions and the resulting traffic light outputs).

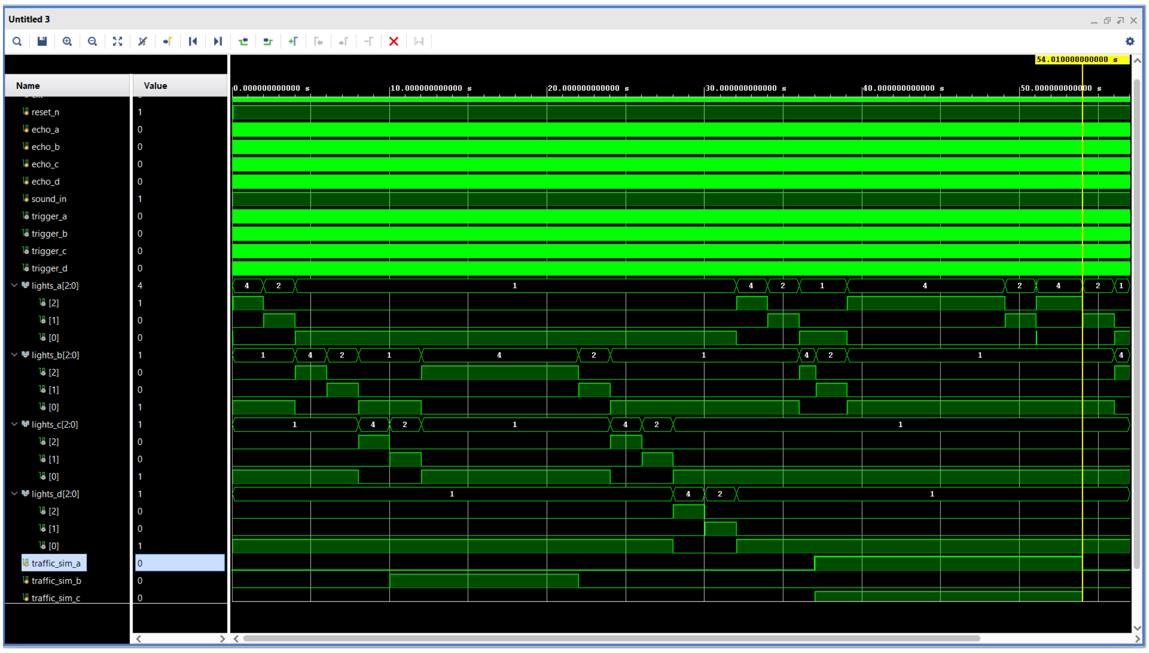

Figure 7.7: Final combined traffic_fsm simulation waveform showing: (top) clock/reset, (middle) major FSM state labels and timers, (bottom) sensor inputs (echo/sound) and lane light outputs. This waveform integrates and highlights how sensor events map to FSM decisions and outputs across multiple cycles.

Key Takeaways :

Figure 6.2 (main FSM diagram) and the combined simulation (Figure 7.7) together show the decision flow:

- The system rests in `DECIDE_NEXT`, where the FSM evaluates the `soundsensor` (highest priority) and the four `traffic_detected` signals.
- Priority resolution uses the deterministic order: `A` → `B` → `C` → `D`. When multiple lanes assert detection simultaneously, the FSM chooses the highest in this order.
- Emergency interrupts (sound sensor) immediately route the FSM to the corresponding lane's GREEN state (pre-empting others). Lane A is designated highest and cannot be interrupted.
- GREEN → YELLOW transitions are timer-driven (GREENTIME then YELLOWTIME). After YELLOW the FSM returns to `DECIDE_NEXT` for re-evaluation.
- Extended green logic: if a lane's sensor remains active, the FSM may extend GREEN up to a configured maximum (MAX_GREEN_TIME) avoiding starvation while giving preference to lanes with continued traffic.

These diagrams and the final waveform confirm the FSM behaves deterministically and robustly under all simulated scenarios.

Chapter 8

Future Scope

Potential extensions include:

- **Fusion with camera or radar sensors** for vehicle classification and speed estimation.
- **Deployment of machine-learning models** (e.g., LSTMs, reinforcement learning) for predictive signal control.
- **Networking multiple junctions via IoT** for city-wide coordination and green corridors.
- **Integration of pedestrian detection and red-light violation capture** for enhanced safety.
- **Advanced acoustic processing** to distinguish types and directions of emergency vehicles.

References

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